Answers learning task 4

2026 – 2027

Exercise 3.79: Let X be a binomial random variable with $p = 0.1$ and $n = 10$. Calculate the following probabilities from the binomial probability mass function.

(a) $P(X \leq 2)$
(b) $P(X > 8)$
(c) $P(X = 5)$
(d) $P(5 \leq X \leq 7)$

Answer

We can use the graphical calculator to calculate the following probabilities:

(a) $P(X \leq 2) = \text{binomcdf}(n = 10, p = 0.1, k = 2) \approx 0.9298$.

(b) $P(X > 8) = 1 - P(X \leq 8) = 1 - \text{binomcdf}(n = 10, p = 0.1, k = 8) \approx 9.1 \cdot 10^{-9}$.

(c) $P(X = 5) = \text{binompdf}(n = 10, p = 0.1, k = 5) \approx 0.0015$.

(d)

$$\begin{aligned} P(5 \leq X \leq 7) &= P(X \leq 7) - P(X \leq 4) \\ &= \text{binomcdf}(n = 10, p = 0.1, k = 7) - \text{binomcdf}(n = 10, p = 0.1, k = 4) \\ &\approx 0.0016. \end{aligned}$$

Exercise 3.91: An article in *Information Security Technical Report* "Malicious Software—Past, Present and Future" (2004, Vol. 9, pp. 6–18) provided the following data on the top 10 malicious software instances for 2002. The clear leader in the number of registered incidences for the year 2002 was the Internet worm "Klez", and it is still one of the most widespread threats. This virus was first detected on 26 October 2001, and it has held the top spot among malicious software for the longest period in the history of virology.

The 10 most widespread malicious programs for 2002

Place	**Name**	**% Instances**
1	I-Worm.Klez	61.22 %
2	I-Worm.Lentin	20.52 %
3	I-Worm.Tanatos	2.09 %
4	I-Worm.BadtransII	1.31 %
5	Macro.Word97.Thus	1.19 %
6	I-Worm.Hybris	0.60 %
7	I-Worm.Bridex	0.32 %
8	I-Worm.Magistr	0.30 %
9	Win95.CIH	0.27 %
10	I-Worm.Sircam	0.24 %

(Source: Kaspersky Labs).

Suppose that 20 malicious software instances are reported. Assume that the malicious sources can be assumed to be independent.

(a) What is the probability that at least 3 instances is "Klez"?

(b) What is the probability that 7 or more instances are "Klez"?

(c) What are the mean and standard deviation of the number of "Klez" instances among the 20 reported?

Answer

Let X be the random variable counting the number of reported instances that are "Klez". Then X follows a binomial distribution with $n = 20$ and $p = 0.6122$. We can use the graphical calculator to calculate the following probabilities:

(a) $P(X \geq 3) = 1 - P(X \leq 2) = 1 - \text{binomcdf}(n = 20, p = 0.6122, k = 2) \approx 0.999997$.

(b) $P(X \geq 7) = 1 - P(X \leq 6) = 1 - \text{binomcdf}(n = 20, p = 0.6122, k = 6) \approx 0.9953$.

(c) The mean and standard deviation of a binomial distribution with parameters n and p can be computed as follows:

$$E(X) = n \cdot p = 20 \cdot 0.6122 = 12.244.$$
$$\sigma(X) = \sqrt{n \cot p \cdot (1 - p)} = \sqrt{20 \cdot 0.6122 \cdot (1 - 0.6122)} \approx 2.1790.$$

Exercise 3.95: Because not all airline passengers show up for their reserved seat, an airline sells 130 tickets for a flight that holds only 120 passengers. The probability that a passenger does not show up is 0.10, and the passengers behave independently.

(a) What is the probability that every passenger who shows up can take the flight?

(b) What is the probability that the flight departs with empty seats?

Answer

Let X be the random variable counting the number of passengers showing up for their reserved seat. Then X follows a binomial distribution with $n = 130$ and $p = 0.90$. We can use the graphical calculator to calculate the following probabilities:

(a) The probability that every passenger who shows up can take the flight is the probability that at most 120 people show up:

$$P(X \leq 120) = \text{binomcdf}(n = 130, p = 0.90, k = 120) \approx 0.8479.$$

(b) The probability that the flight departs with empty seat is the probability that strictly fewer than 120 people show up:

$$P(X < 120) = P(X \leq 119) = \text{binomcdf}(n = 130, p = 0.90, k = 119) \approx 0.7619.$$

Exercise 3.103: Consider a sequence of independent Bernoulli trials with $p = 0.25$.

(a) What is the expected number of trials to obtain the first success?

(b) After the eighth success occurs, what is the expected number of trials to obtain the ninth success?

Answer

(a) The number of trials Y to obtain the first success is a geometrically distributed random variable with parameter $p = 0.25$. The expected value $E(Y) = \frac{1}{p} = \frac{1}{0.25} = 4$.

(b) Since each of the trials is independent of each other, the information of past trials does not change the probabilities and hence expected value of the number of trials to obtain the next future success. Therefore, this expected number of trials is once again equal to 4.

Exercise 3.105: The probability of a successful optical alignment in the assembly of an optical data storage product is 0.7. Assume that the trials are independent.

(a) What is the probability that the first successful alignment requires exactly four trials?

(b) What is the probability that the first successful alignment requires at most four trials?

(c) What is the probability that the first successful alignment requires at least four trials?

Answer

The number of trials X for a succesful optical alignment is a geometric random variable with $p = 0.7$. We can compute the following probabilities using the graphical calculator:

(a) $P(X = 4) = \text{geometpdf}(p = 0.7, k = 4) = 0.0189$.

(b) $P(X \leq 4) = \text{geometcdf}(p = 0.7, k = 4) = 0.9919$.

(c) $P(X \geq 4) = 1 - P(X \leq 3) = 1 - \text{geometcdf}(p = 0.7, k = 3) = 0.027$.

Exercise 3.131: Suppose that X has a Poisson distribution with a mean of 5. Determine the following probabilities:

(a) $P(X = 0)$
(b) $P(X \leq 2)$
(c) $P(X = 4)$
(d) $P(X = 8)$

Answer

We can use the graphical calculator to compute this probabilities:

(a) $P(X = 0) = \text{poissonpdf}(\mu = 5, k = 0) \approx 0.0067$.

(b) $P(X \leq 2) = \text{poissoncdf}(\mu = 5, k = 2) \approx 0.1247$.

(c) $P(X = 4) = \text{poissonpdf}(\mu = 5, k = 4) \approx 0.1755$.

(d) $P(X = 8) = \text{poissonpdf}(\mu = 5, k = 8) \approx 0.0653$.

Exercise 3.133: Suppose that the number of customers who enter a bank in an hour is a Poisson random variable, and suppose that $P(X = 0) = 0.1$. Determine the mean and variance of X.

Answer

Notice that for a Poisson random variable X, we have that

$$P(X = 0) = e^{-\mu} \cdot \frac{\mu^0}{0!} = e^{-\mu}.$$

As $P(X = 0) = 0.1$, we have that $e^{-\mu} = 0.1$, or $-\mu = \ln 0.1 \Rightarrow \mu = \ln 10$. The mean and variance of a random variable $X \sim \text{Poisson}(\mu)$ is equal to μ and μ respectively. Therefore $E(X) = V(X) = \ln 10 \approx 2.3025$.

Exercise 3.134: The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 8 calls per hour.

(a) What is the probability that there are exactly five calls in one hour?

(b) What is the probability that there are three or fewer calls in one hour?

(c) What is the probability that there are exactly 15 calls in two hours?

(d) What is the probability that there are exactly five calls in 30 minutes?

Answer

The number of telephone calls X is Poisson distributed with mean $\mu = 8$ per hour, so $\mu = 16$ per two hours and $\mu = 4$ per 30 minutes. We can calculate the probabilities using the graphical calculator:

(a) $P(X = 5 \mid \mu = 8) = \text{poissonpdf}(\mu = 8, k = 5) \approx 0.0916$,

(b) $P(X \leq 3 \mid \mu = 8) = \text{poissoncdf}(\mu = 8, k = 3) \approx 0.0424$,

(c) $P(X = 15 \mid \mu = 16) = \text{poissonpdf}(\mu = 16, k = 15) \approx 0.0992$,

(d) $P(X = 5 \mid \mu = 4) = \text{poissonpdf}(\mu = 4, k = 5) \approx 0.1563$.

Exercise 3.141: The number of surface flaws in plastic panels used in the interior of automobiles has a Poisson distribution with a mean of 0.5 per square meter of plastic panel. Assume an automobile interior contains 1 square meter of plastic panel.

(a) What is the probability that there are no surface flaws in an auto's interior?

(b) If 10 cars are sold to a rental company, what is the probability that none of the 10 cars has any surface flaws?

(c) If 10 cars are sold to a rental company, what is the probability that at most one car has any surface flaws?

Answer

The number of surface flaws X is a Poisson random variable with $\mu = 0.5$ per square meter (or per car). Therefore, the mean $\mu = 5$ per 10 cars. We can calculate the probabilities using the graphical calculator:

(a) $P(X = 0 \mid \mu = 0.5) = \text{poissonpdf}(\mu = 0.5, k = 0) \approx 0.6065$,

(b) $P(X = 0 \mid \mu = 5) = \text{poissonpdf}(\mu = 8, k = 3) \approx 0.0067$,

(c) $P(X \leq 1 \mid \mu = 5) = \text{poissoncdf}(\mu = 4, k = 1) \approx 0.0404$.

Exercise 3.171: Determine the probability mass function for the random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < 2 \\ 0.35 & 2 \leq x < 5.7 \\ 0.5 & 5.7 \leq x < 6.5 \\ 0.7 & 6.5 \leq x < 8.5 \\ 1 & 8.5 \leq x \end{cases}$$

Answer

In order to determine the probability mass function for the random variable with this CDF, we need to consider the boundary values for the cases in the CDF, so $x = 2$, $x = 5.7$, $x = 6.5$ and $x = 8.5$, and see how large the difference is for the two cases they are involved in. This gives us the following probability mass function:

$$f(x) = \begin{cases} 0.35 & \text{for } x = 2 \\ 0.15 & \text{for } x = 5.7 \\ 0.20 & \text{for } x = 6.5 \\ 0.30 & \text{for } x = 8.5 \end{cases}$$